Plane mirrors

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use the reflection law and locate a virtual image.

SUCCESS CRITERIA - I can...

1. Measure angles from the normal.
2. Use equal incidence and reflection angles.
3. Place the image the same perpendicular distance behind the mirror.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A wall is **50 m** away. An echo returns after **0.5 s**. A student divides one-way distance by this time. **Find the correct speed** and **explain the student's error**.

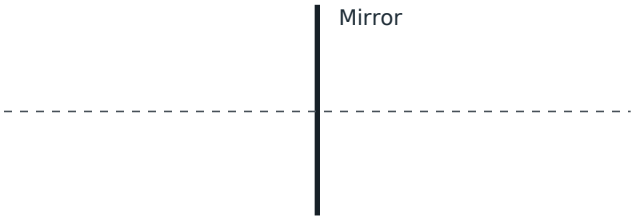
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\theta_{incident} = \theta_{reflected}$
image distance behind = object distance in front.
Plane mirror; **measure angles** from the normal. The image is virtual.

Worked reasoning: An object **2 m** in front of an ideal plane mirror has a virtual image **2 m** behind it. The object-image separation is **4 m**; reflected rays appear to come from the image.

Your turn: Draw the mirror, object, image and backward extensions of two reflected rays.

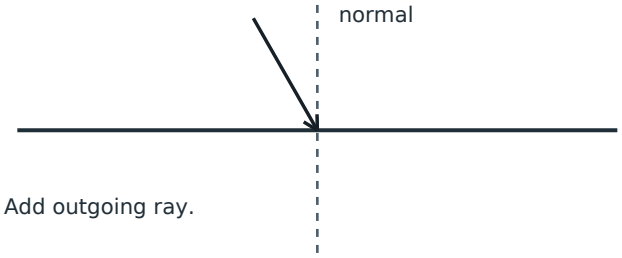


Add object / image or incident / reflected rays.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Draw the normal at the mirror. **Complete a reflected ray** for an incident ray **30°** from normal.

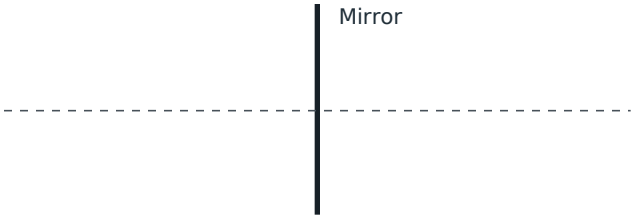


Add outgoing ray.

Marzano check - circle: ① ② ③

04 Apply the physics

A small lamp is **2 m** in front of a plane mirror. **Mark the lamp** and its virtual image on the diagram. **Find image distance behind the** mirror and the full lamp-to-image separation.



Add object / image or incident / reflected rays.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Why is the mirror image called virtual rather than a real object behind the glass?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

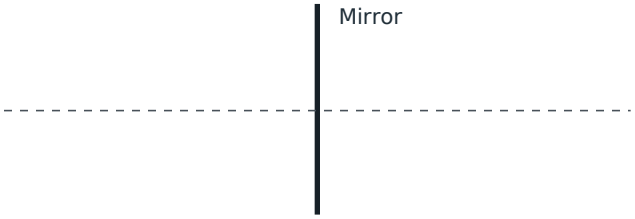
A reflected signal returns after **4 s**. **Name the extra quantity needed** to **calculate distance** and **explain why this** is a round trip.
Label the outgoing and returning paths.



Marzano check - circle: 1 2 3

07 Exit check

Incidence angle **45°** from normal. **Find reflection angle**.

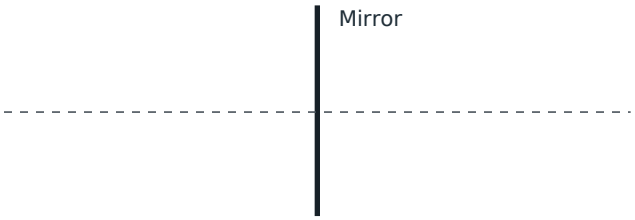


Add object / image or incident / reflected rays.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Move the object **0.5 m** closer to the mirror. **Find the new image distance** and object-image separation.



Add object / image or incident / reflected rays.

Marzano check - circle: 1 2 3